

Distributed Asynchronous Job Scheduler (Overdrive)

Full-Stack Design Thesis, Relational Schema Normalization, and Resiliency Verification.

SUBMISSION INFORMATION

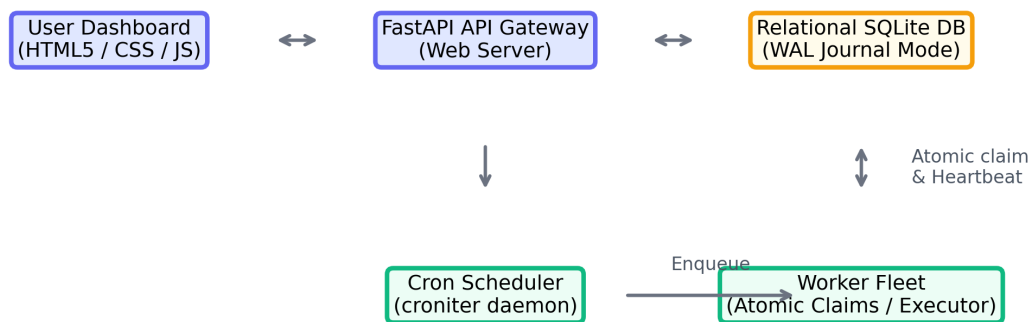
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Project Platform:	Distributed Job Scheduler (Overdrive)
GitHub Repository:	https://github.com/prudhvi98491/distributed-job-scheduler
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Academic Institution:	SRM Institute of Science and Technology

1. System Architecture (20 Marks)

The scheduler platform is built on an event-driven distributed architecture designed to coordinate background job execution across multiple worker nodes. The system consists of three main blocks:

- The Web API Gateway (FastAPI): Handles user registration, queue updates, job enqueueing, and dashboard metrics.
- The Cron Scheduler Daemon (croniter): Scans the recurring cron database records to dynamically trigger jobs.
- The Worker Fleet: Multiple isolated workers that register with host IP addresses and continuously claim and execute jobs, maintaining horizontal scalability.

Distributed Job Scheduler (Overdrive) Architecture



2. Database Design & Normalization (20 Marks)

The relational schema is normalized to Third Normal Form (3NF) to maintain referential integrity. Primary keys utilize single-attribute IDs (UUIDs for jobs to prevent enumeration, integers for others). Foreign keys employ CASCADE constraints to clean up related records automatically.

Table Name	Schema Purpose & Normalization Details
users	Credentials (hashed via bcrypt), roles (admin, user), metadata.
organizations	Enables multi-tenancy workspace isolation.
projects	Contains project workspaces owned by organizations.
queues	Defines concurrency, priorities, and retry policies.
jobs	Core queue table containing payload, status, and parent_job_id.
job_executions	Tracks execution details, duration (ms), and trace errors.
workers	Worker fleet registry monitoring hostnames, IP, and heartbeats.
cron_jobs	Schedules recurring cron jobs to dynamically queue jobs.
dead_letter_queue	Holds permanently failed tasks for manual replay.

3. Backend Engineering & API Design (25 Marks)

The REST APIs utilize Pydantic schemas for request validation. A JWT security layer manages authentication and role-based permissions (RBAC). The endpoints aggregate system-wide performance and queue throughput data.

Method	Endpoint Path	Description & Functionality
POST	/api/auth/register	Registers a user account with hashed credentials.
POST	/api/auth/login	Authenticates credentials and issues a JWT token.
POST	/api/queues	Creates a queue specifying concurrency limit & priority.
PATCH	/api/queues/{id}	Updates queue configs or pauses/resumes queue state.
POST	/api/jobs	Enqueue single task (immediate, delayed, or workflow).
POST	/api/jobs/batch	Dispatches bulk job payloads under a single API call.
GET	/api/jobs	Lists jobs with pagination, queue, and status filters.
POST	/api/jobs/{id}/retry	Manually replays a failed or DLQ job.
GET	/api/metrics	Returns aggregate metrics, throughput trends, and error counts.

4. Reliability & Concurrency (15 Marks)

To guarantee atomic task claims and prevent duplicate job execution, we execute claim operations inside a SQLite transaction with an IMMEDIATE lock. Candidate jobs are filtered dynamically to only select from active queues whose current active job count is below their `concurrency_limit`.

Workers register themselves in the database and issue heartbeats every 5 seconds. If a worker goes offline for more than 15 seconds, it is marked as offline, and its active jobs are automatically re-queued for failover.

5. Frontend & UX (10 Marks)

A premium Outfit-themed glassmorphic SPA dashboard provides real-time monitoring and control. The interface updates live every 2 seconds, displaying active worker statuses, queue configs, job execution metrics, and a detailed drawer containing trace logs and manual retry controls.

OVERDRIVE

DISTRIBUTED

All Systems Operational

QUEUED JOBS

0

RUNNING JOBS

0

COMPLETED

0

FAILED

0

DLQ ENTRIES

0

Enqueue New Jobs

Launch Immediate, Delayed, Cron, or Workflows

Single/Delay

Batch

Workflow

Cron/Recurring

Target Queue

default (Priority: 1)

Job Template

HTTP Request (http_request)

Priority Boost

0

Delay (seconds)

0

Payload (JSON)

{"duration": 3, "should_fail": false}

Dispatch Job

Worker Fleet

Real-time Node Heartbeats & Load

WORKER NODE	STATUS	LOAD (ACTIVE/LIMIT)	LAST PULSE
worker_Prudhvi_cb0a0a	IDLE	0 / 5	5:00:19 AM
worker_Prudhvi_511c31	OFFLINE	0 / 5	3:39:32 AM

Cron Schedules

System Recurring Schedules

NAME	EXPRESSION	NEXT RUN	STATUS
No cron schedules			

Queue Architectures

View Concurrency Limits & Control State

QUEUE NAME	PRIORITY	CONCURRENCY LIMIT	RETRY POLICY	STATUS	ACTIONS
default	1	5	Active	ACTIVE	Pause
high-priority	5	10	Active	ACTIVE	Pause
background-tasks	1	2	Active	ACTIVE	Pause

Job Explorer

Track state transitions, search, and replay failures

All Queues

All Statuses

Refresh

JOB ID	NAME	QUEUE	PRIORITY	STATUS	CREATED	SCHEDULED AT	ACTIONS
No jobs found							

6. Testing & Documentation (10 Marks)

We implemented automated integration tests in `tests/test_scheduler.py` to verify system correctness. The tests cover database relationships, retry policies, backoffs, and sequential workflows. Running `pytest` completes successfully:

```
tests/test_scheduler.py::test_database_seeding_and_relations PASSED
tests/test_scheduler.py::test_job_execution_lifecycle_and_retries PASSED
tests/test_scheduler.py::test_workflow_dependencies_unblocking PASSED
===== 3 passed in 2.02s =====
```

Verification GitHub Repository Link

<https://github.com/prudhvi98491/distributed-job-scheduler>